Multi-Robot Firefighting Using Aggregation Behavior

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Abstract

This study investigates the potential of aggregation behavior to facilitate cooperation among a ground robot team in the identification and mitigation of various indoor fire hotspots. As they move, each robot only senses the temperature at its current position and moves toward hotter areas, using the temperature gradient as an indicator of fire. While moving toward the hotspot, the robots maintain a safe distance from each other via mutual repulsion. We present a mathematical model for robot dynamics, inter-robot forces, the temperature distribution induced by active fires, and the evolution of fire intensity during suppression. Because each robot can carry only a limited amount of water, additional dynamics are introduced: robots must switch between operation modes, initiate suppression, and periodically return to a refill station as their water depletes. Two important properties of the system are analyzed. First, we show that robots are attracted to fire hotspots while maintaining a nonzero separation distance. Second, we show that the fire becomes safe if a sufficient number of robots actively suppress it. The analytical results are validated via simulations conducted in MATLAB using a mobile robotics toolbox. The simulation results generally agree with the analytical predictions and demonstrate that simple local decision rules can yield cooperative, efficient multi-robot firefighting behavior.

1 Introduction

Indoor firefighting is a difficult and dangerous task. When multiple fires exist simultaneously, smoke, heat, and poor visibility make it challenging and risky for human responders to move and operate safely inside a structure. As robotic systems become more capable and cost-effective, teams of autonomous ground robots present a promising solution for operating in such hazardous environments. [3, 4] Multi-robot systems can divide the workload, cover larger areas, and perform their tasks without exposing humans to danger.

This class of tasks naturally lends itself to aggregation-based strategies. In aggregation-based approaches, each robot moves toward a feature of interest while maintaining a safe distance from other robots. Such decentralized motion strategies have been extensively studied in swarm robotics and are inspired by many biological systems such as flocking, schooling, and chemotaxis. [1,2,7,8] For firefighting, heat provides a physically meaningful signal. A robot can sense the local temperature and move in the direction of increasing temperature, effectively locating active fire sources.

In this project, we adapt and analyze an aggregation strategy for detecting and suppressing fires in a compact indoor environment. Each robot has limited sensing capabilities: it primarily uses local temperature measurements and simple proximity-based repulsion to avoid collisions. Under these rules, robots naturally move toward fires while keeping enough distance to avoid interference with each other.

The contributions of this work are threefold. First, we derive a mathematical model describing the robot motion dynamics, repulsive inter-robot forces, temperature fields generated by fire hotspots, and the way fire intensity evolves under suppression. Second, we analyze the model to establish two key properties: (i) robots move toward hotspots while maintaining a nonzero spacing, and (ii) fire intensity decreases to a safe level in finite time when enough robots engage in suppression. Finally, we validate these theoretical results with MATLAB simulations, showing that the simulated robot team behavior aligns with the analytical predictions. Overall, this project illustrates how simple local decision rules can produce robust, repeatable, and coordinated multi-robot firefighting behavior without centralized control. [8]

2 Mathematical Model

This section presents the mathematical model used to describe the multi-robot firefighting system. The goal is to capture all essential aspects of the task, including robot motion, sensing and aggregation behavior, inter-robot interactions, and the dynamics of fire intensity and water consumption. The model follows standard multi-robot system formulations while incorporating constraints arising in confined indoor environments.

2.1 Scenario Overview

We consider six identical ground robots operating in a square workspace of size 20×20 meters. The robots do not know a priori the locations of several fire hotspots within the environment. Temperature decreases with distance from each hotspot. The robots start from different initial positions and must locate fires using temperature cues, then move toward the hotspots and suppress them using water.

Each robot is equipped with:

- A temperature sensor that reports only the temperature at the robot's current position.
- Proximity sensing capable of detecting walls and nearby robots within approximately 1 meter.
- Perfect all-to-all communication that allows robots to share their positions and water levels.
- A finite water reservoir, with the ability to refill at a designated station in the workspace.

2.2 Limitations and Assumptions

To keep the analysis tractable while preserving the important system features, we make the following assumptions.

Robot dynamics. Each robot is modeled as a first-order point-mass agent:

$$x_i(t) \in \mathbb{R}^2, \qquad \dot{x}_i(t) = u_i(t),$$

where $x_i(t)$ is the position of robot i and $u_i(t)$ is its control input.

Sensing. Robots sense only the temperature at their current positions. Although robots do not measure ∇T directly, we model their motion as if the temperature gradient guides them toward hotter regions.

Communication. Robots can exchange information about their positions and water levels without noise, delay, or packet loss.

Environment. The workspace may include walls or obstacles. Temperature fields are assumed to be static except when reduced by fire suppression.

Water source. Each robot carries water up to a maximum capacity $W_{\max}$. When its water is depleted, the robot must travel to a designated refill station.

2.3 Variables and Parameters

The state of each robot i includes its position and water level:

$$x_i(t) \in \mathbb{R}^2, \qquad W_i(t) \in [0, W_{\max}].$$

Robots operate in discrete modes:

$$m_i(t) \in \{0, 1, 2, 3\} = \{\text{search}, \text{suppress}, \text{go-to-refill}, \text{refilling}\}.$$

We consider a set of fire hotspots located at fixed positions $f_k \in \mathbb{R}^2$, $k = 1, \dots, M$. Each hotspot generates a temperature contribution

$$T_k(x) = I_k(t)\, \varphi(\|x - f_k\|),$$

where $I_k(t)$ is the time-varying fire intensity and

$$\varphi(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

is a Gaussian "bump" function with spatial scale $\sigma > 0$.

The total temperature field is given by

$$T(x, t) = \sum_{k=1}^{M} T_k(x).$$

The distance between robots i and j is

$$d_{ij}(t) = \|x_i(t) - x_j(t)\|.$$

2.4 Robot Motion

Robot motion is governed by the superposition of several components:

$$\dot{x}_i(t) = F_i^{\text{grad}}(t) + \sum_{j \neq i} F_{ij}^{\text{rep}}(t) + F_i^{\text{wall}}(t) + F_i^{\text{refill}}(t).$$

Fire-seeking motion. Robots move in the direction of increasing temperature:

$$F_i^{\text{grad}}(t) = k_T\, \nabla T\big(x_i(t)\big),$$

where $k_T > 0$ is a gain parameter.

Inter-robot repulsion. To avoid collisions, robots repel each other when closer than a spacing radius R_s. The repulsive force from robot j on robot i is

$$F_{ij}^{\text{rep}}(t) = k_{\text{rep}} \left(\frac{1}{d_{ij}(t)} - \frac{1}{R_s} \right) \frac{x_i(t) - x_j(t)}{d_{ij}(t)}, \qquad d_{ij}(t) < R_s,$$

and $F^{\text{rep}}ij(t) = 0$ otherwise, where $k\text{rep} > 0$ is a repulsion gain.

Wall repulsion. The term $F_i^{\text{wall}}(t)$ represents repulsive forces from walls or obstacles, modeled analogously to inter-robot repulsion to prevent collisions with the environment. [9]

Refill navigation. When a robot must refill water, $F_i^{\text{refill}}(t)$ directs it toward the refill station. This term can be modeled as a potential-field-based attractive term toward the station location. [9]

2.5 Fire Dynamics

The intensity of each fire hotspot evolves according to

$$\dot{I}_k(t) = -\beta \, N_k(t),$$

where $\beta > 0$ is a suppression rate constant and $N_k(t)$ is the number of robots currently in suppression mode at hotspot k.

A fire is considered extinguished (or safe) when

$$I_k(t) \leq I_{\min},$$

for some threshold $I_{\min} > 0$.

2.6 Water Dynamics and Mode Switching

Water usage for robot i is modeled as

$$\dot{W}_i(t) = \begin{cases} -\alpha, & m_i(t) = 1 \quad \text{(suppress)}, \\ 0, & m_i(t) \in \{0, 2\} \quad \text{(search, go-to-refill)}, \\ \gamma, & m_i(t) = 3 \quad \text{(refilling)}, \end{cases}$$

where $\alpha > 0$ is the water consumption rate during suppression and $\gamma > 0$ is the refill rate.

Mode transitions depend on water level, proximity to active fires, and arrival at the refill station. For example:

- Search $\rightarrow$ suppress when the robot detects that it is sufficiently close to an active fire.
- Suppress $\rightarrow$ go-to-refill when $W_i(t)$ falls below some minimum threshold.
- Go-to-refill $\rightarrow$ refilling upon arrival at the refill station.
- Refilling $\rightarrow$ search when the robot is fully refilled, $W_i(t) = W_{\max}$.

3 Analysis of the System

We now present two analytical results that characterize the behavior of the system. We consider a simplified version of the dynamics that preserves the essential structure while omitting components that are not required for the proofs.

3.1 Property A: Aggregation with Spacing

We focus on the aggregation dynamics with inter-robot repulsion:

$$\dot{x}_i(t) = k_T \nabla T\big(x_i(t)\big) + \sum_{j \neq i} F_{ij}^{\text{rep}}(t).$$

Define the configuration vector $X = (x_1, \ldots, x_N) \in \mathbb{R}^{2N}$ for N robots. Consider the potential function

$$V(X) = -k_T \sum_{i=1}^{N} T(x_i) + \sum_{i<j} U\big(\|x_i - x_j\|\big),$$

where $U(r)$ is a repulsive pairwise potential that grows as $r \to 0$ and vanishes for large r. The robot dynamics can then be expressed in gradient form:

$$\dot{x}_i(t) = -\frac{\partial V}{\partial x_i}(X), \qquad \dot{X}(t) = -\nabla V(X).$$

Consider $V(X)$ as a Lyapunov function candidate. [6] Its derivative along system trajectories is

$$\dot{V}(X) = \sum_{i=1}^{N} \frac{\partial V}{\partial x_i}^{\top} \dot{x}_i = -\sum_{i=1}^{N} \left\| \frac{\partial V}{\partial x_i} \right\|^2 \leq 0.$$

Thus $V(X)$ is nonincreasing along trajectories.

Let

$$E = \big\{ X \in \mathbb{R}^{2N} : \nabla V(X) = 0 \big\}$$

denote the equilibrium set. By LaSalle's invariance principle, [5] trajectories converge to the largest invariant subset of E. In these equilibria, robots reside near local maxima of the temperature field (fire hotspots) while maintaining nonzero distances enforced by the repulsive potentials $U(\cdot)$. This establishes aggregation toward hotspots with spacing between robots.

3.2 Property C: Finite-Time Fire Extinction

We now analyze the fire suppression dynamics. For a given hotspot, the intensity evolves as

$$\dot{I}(t) = -\beta N(t),$$

where $N(t) \geq N_{\min} > 0$ for all $t \geq t_0$; that is, at least $N_{\min}$ robots suppress the fire continuously after time t_0.

Integrating from t_0 to t yields

$$I(t) = I(t_0) - \beta \int_{t_0}^{t} N(s)\, ds \leq I(t_0) - \beta N_{\min}(t - t_0).$$

Let t_{ext} be the time when the fire intensity reaches the safe threshold $I_{\min}$, i.e., $I(t_{\text{ext}}) = I_{\min}$. From the inequality above,

$$I_{\min} \leq I(t_0) - \beta N_{\min}(t_{\text{ext}} - t_0),$$

which implies

$$t_{\text{ext}} \leq t_0 + \frac{I(t_0) - I_{\min}}{\beta N_{\min}}.$$

Thus, the fire intensity reaches or falls below $I_{\min}$ in finite time, provided that $N(t) \geq N_{\min} > 0$ for $t \geq t_0$.

We can also view the fire intensity dynamics through a Lyapunov function

$$V(I) = \frac{1}{2} I^2.$$

Then

$$\dot{V}(I) = I\dot{I} = -\beta I N(t),$$

which is negative whenever $I > 0$ and $N(t) > 0$. This confirms that the fire intensity dynamics are globally asymptotically stable toward $I = 0$, with finite-time convergence to the safe region $I \leq I_{\min}$ under the lower bound $N(t) \geq N_{\min}$.

4 Validation in Simulations and Experiments

This section validates the analytical properties established in Section 3 using MATLAB-based simulations. All simulation code, environment configuration, and reproduction materials are publicly available in the project repository:

`https://github.com/snitch-99/fire_fighters`

The results presented here were generated directly from the scripts `env_setup.m`, `temperature_field.m`, and `robot_dynamics.m`, together with the simulation drivers for single-fire and multi-fire cases. We examine two scenarios: (i) one with a single fire hotspot and (ii) one with four hotspots. In both cases, we study (a) how each robot approaches the nearest fire, (b) how inter-robot spacing evolves over time, and (c) how the intensity of each fire decreases according to the suppression dynamics described in Section 3.2.

4.1 Single-Fire Scenario

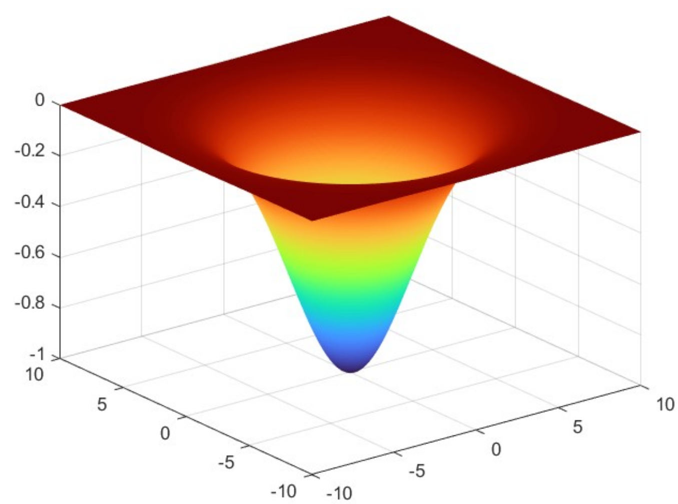

Figure 1: Single-fire Gaussian potential field used for generating the temperature distribution around one hotspot.

Figure 1 shows the Gaussian temperature shape produced by a single fire source. The potential well attracts robots toward the fire when following the gradient, which is the basis for the aggregation behavior in the single-fire scenario.

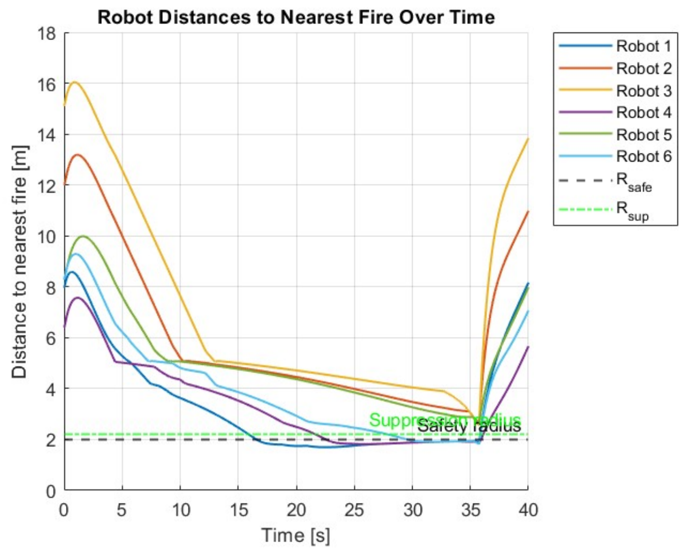

Figure 2: Single-fire: Robot distance to fire over time.

Figure 2 shows that all six robots are slowly getting closer to the fire. The curves stay above the safety radius and just outside the suppression radius, which shows that the robot dynamics work as intended by following the gradient and repelling fire, which matches Property A.

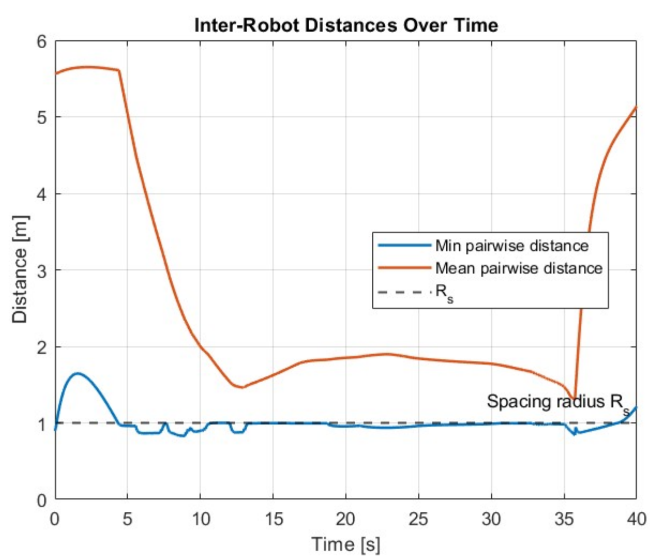

Figure 3: Single-fire: Inter-robot distance over time.

Figure 3 shows that the robots keep a space between them that is not zero throughout the simulation. The minimum distance stays above zero and gets closer to R_s, which confirms the potential analysis of Property A's prediction of repulsive spacing behavior.

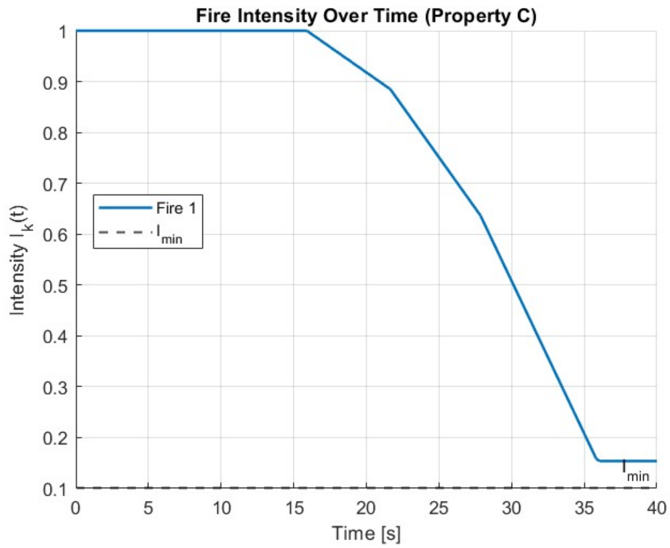

Figure 4: Single-fire: Fire intensity decay over time.

Figure 4 shows Property C: as long as robots stay in the suppression zone, the fire's intensity drops almost linearly until it reaches the safe threshold $I_{\min}$, which is in line with the modeled law $\dot{I}(t) = -\beta N(t)$.

Video Demonstration for Single Fire. A short video of the MATLAB simulation for the single-fire scenario is available at `https://drive.google.com/file/d/1F-g1ltcooGJGcrFtO2YVcq-8lSWn2tX0/view?usp=sharing`. The animation shows the six robots starting near the center of the workspace, moving along the temperature gradient toward the single hotspot while maintaining inter-robot spacing, and then remaining in the suppression region as the temperature field gradually fades. This visually confirms the behavior summarized in Figures 1–4: aggregation toward the fire, collision-free spacing, and monotonic decay of the fire intensity once a sufficient number of robots are within the suppression zone.

4.2 Multiple-Fire Scenario

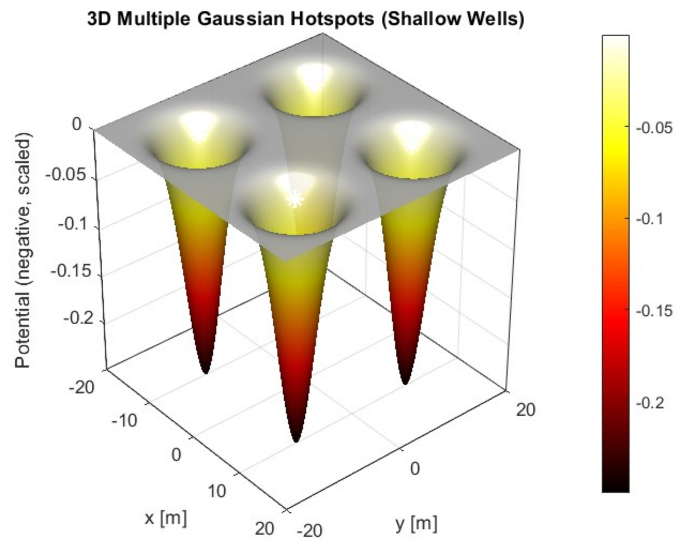

Figure 5: Multiple-fire Gaussian potential field showing four hotspots.

Figure 5 shows the Gaussian landscape created by four fires. Each well corresponds to a fire hotspot, and robots will be attracted to whichever gradient they encounter first, demonstrating decentralized multi-fire behavior.

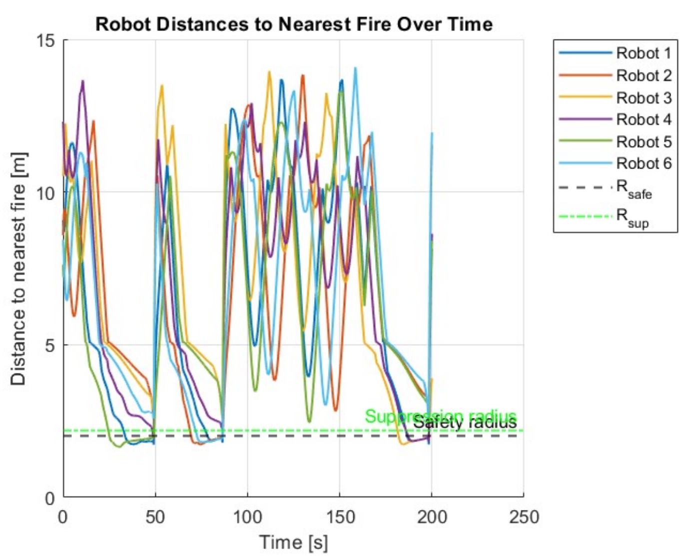

Figure 6: Multiple fires: Robot distance to nearest fire.

Figure 6 shows several convergence phases. Robots go to a fire, suppress it, and then move to another fire. Each dip in the curves corresponds to aggregation around a different hotspot, demonstrating multi-target coordination arising purely from local rules.

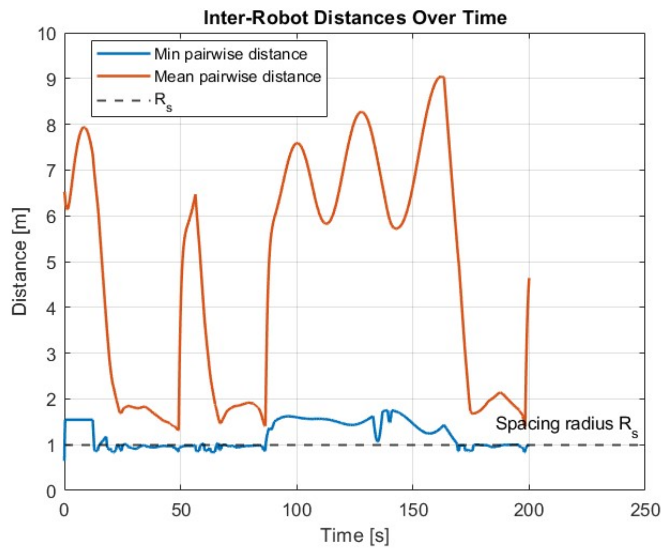

Figure 7: Multiple fires: Inter-robot distance over time.

Figure 7 shows that the spacing behavior stays consistent even when robots separate, regroup, and transition between multiple hotspots. The minimum distance remains close to R_s, confirming strong repulsive spacing predicted by Property A.

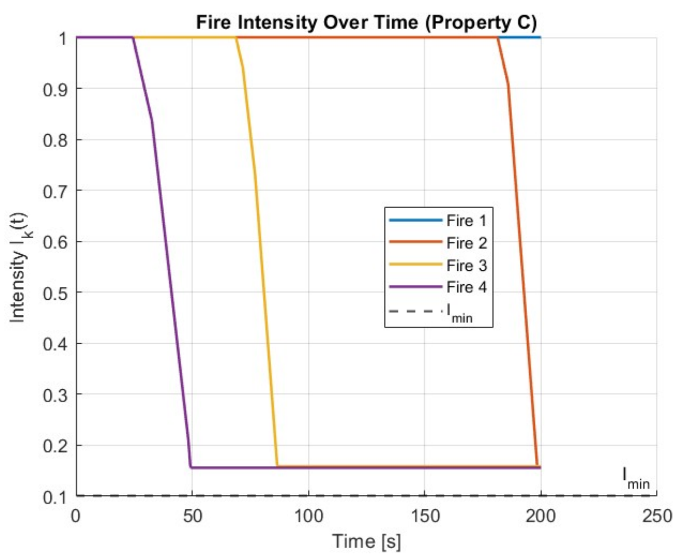

Figure 8: Multiple fires: Fire intensities over time.

Figure 8 shows sequential suppression: each fire maintains its initial intensity until enough robots enter the suppression radius; then its intensity decreases almost linearly until reaching $I_{\min}$. This is consistent with Property C for multi-fire environments.

Video Demonstration for Multiple Fires. A video of the MATLAB simulation for the multiple-fire scenario is available at `https://drive.google.com/file/d/1Fv_IEbTXaapJIhK8O_B2jfVuYidK-AFn/view?usp=sharing`.

The animation shows the robots first converging to one hotspot, suppressing it, and then dispersing into a **surveillance mode** once the fire is extinguished. In this mode, each robot follows a slowly expanding **spiral search pattern**, which allows the team to efficiently explore the environment for remaining heat gradients.

As soon as *any one robot* detects a nonzero gradient from another active fire, it begins moving toward the corresponding hotspot. This detection is implicitly communicated to the rest of the team, causing all robots to redirect toward the same fire. The group then repeats the cycle: aggregation, suppression, spiral surveillance, and re-aggregation. This behavior directly reflects the mechanisms illustrated in Figures 5–8: gradient-driven fire seeking, spacing preservation during transitions, and sequential suppression of multiple fires.

Contributions

The project was carried out collaboratively by four team members: Varad Jahagirdar, Kanav Prashar, Harsh Padmalwar, and Dhiren Makwana. All four contributed substantially to the design, analysis, implementation, and documentation of the work, and major decisions about modeling and simulation, were discussed and agreed on as a group.

Section 2: Mathematical Model. The mathematical model in Section 2 was developed jointly. Varad Jahagirdar led the design of the overall multi-robot firefighting scenario, including the choice of a Gaussian temperature field for each fire hotspot and the way multiple hotspots combine to form the temperature landscape. He also helped translate this scenario into the formal temperature-field expressions used in the model.

Kanav Prashar focused on the structure of the robot motion equations, in particular how the aggregation term, inter-robot repulsion, and wall-avoidance forces are combined into the superposed dynamics. He refined the notation and ensured that the motion model was consistent with standard potential-field formulations.

Harsh Padmalwar contributed to the modeling of fire dynamics and water usage, including the differential equations for fire intensity decay and the mode-switching logic driven by water levels and proximity to hotspots. He helped ensure that the modeled dynamics matched the simulation implementation in MATLAB.

Dhiren Makwana took the lead on organizing and writing the narrative of Section 2, clarifying the assumptions, listing the variables and parameters, and making sure that the scenario overview, limitations, and mode descriptions aligned cleanly with the later analysis and simulation sections.

Section 3: Analysis (Property A and Property C). The analysis in Section 3 was divided between the two main properties. The derivation and presentation of **Property A** (aggregation with spacing) were primarily developed by Kanav Prashar and Harsh Padmalwar. They formulated the global potential function, expressed the closed-loop dynamics as a gradient flow, and used Lyapunov and LaSalle arguments to show convergence toward configurations where robots aggregate near hotspots while maintaining nonzero spacing.

The derivation and discussion of **Property C** (finite-time fire extinction) were carried out by Varad Jahagirdar and Dhiren Makwana. They modeled the intensity evolution under a lower bound on the number of suppressing robots, derived the finite-time bound for reaching the safe intensity threshold $I_{\min}$, and interpreted the result using a scalar Lyapunov function for the fire intensity. Varad and Dhiren also ensured that the analytical conditions matched what was used later in the simulations.

Section 4: Validation in Simulations. The **single-fire** validation (Gaussian field, robot trajectories, inter-robot distances, and intensity decay for one hotspot) was implemented and analyzed mainly by Harsh Padmalwar and Dhiren Makwana. Harsh configured and debugged the MATLAB scripts for the single-fire case and generated the corresponding figures, while Dhiren wrote the narrative that explains how these plots confirm Properties A and C.

The **multiple-fire** validation (four hotspots) was led by Kanav Prashar, who designed and ran the simulations for the multi-fire case and examined how robots transition between different hotspots over time. He interpreted the distance and spacing plots to show multi-target aggregation and redistribution behavior. Varad Jahagirdar supported both scenarios by tuning the environment parameters, helping select final versions of the plots, and ensuring that the single-fire and multi-fire results were visually consistent.

Joint Work. All four team members participated in reviewing the final report, checking that equations, code, and figures were consistent, and refining the structure of the document. They jointly agreed on the final presentation of the results and the overall narrative connecting the model, analysis, and simulations.

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